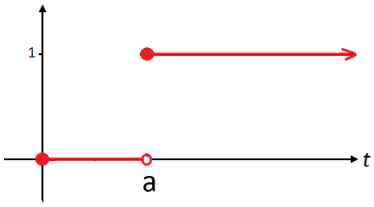
**Definición:** **(Función Escalón Unitaria)**

$$u(t-a)=\begin{cases}0 & , \quad 0\le t< a\\ 1 & , \quad t\ge a\end{cases}$$




**Ejemplo**

$$U(t-2)=\begin{cases}0 & , \quad 0\le t<2\\ 1 & , \quad t\ge 2\end{cases}$$

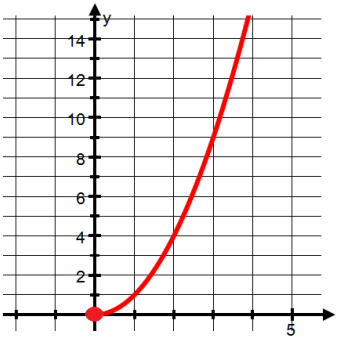
**NOTA:** $u(t-0)=u(t)=1$

**Ejemplo**

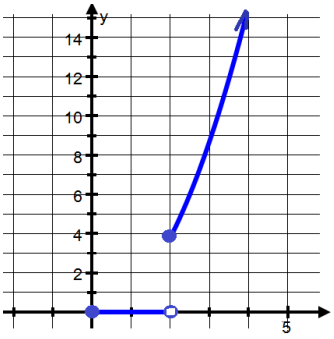
Si $f(t)=t^2$ , t ≥ 0 y $u(t-2)$ Entonces $f(t)u(t-2)=\begin{cases}0 & ,0\le t<2\\ t^2 & ,t\ge 2\end{cases}$

Acá se muestran las gráficas de:

$$f(t)=t^2$$




$$f(t)u(t-2)=\begin{cases}0 & ,0\le t<2\\ t^2 & ,t\ge 2\end{cases}$$




Comentario

Observe que la función escalón se comporta como un “interruptor” que apaga y enciende a la función $f$ .

## Funciones seccionadas expresadas en términos de funciones escalón unitaria

### Ejemplo

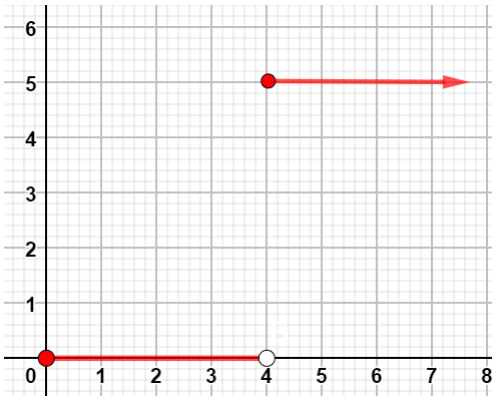
Expresar $f(t)=\begin{cases}5, & 0\le t<2\\ -1, & 2\le t<3\\ 0, & t\ge 3\end{cases}$ en términos de funciones escalón unitarias

Solución

$$f(t)=5\left[u(t-0)-u(t-2)\right]-1\left[u(t-2)-u(t-3)\right]+0\left[u(t-3)\right]$$
$$=5-5u(t-2)-u(t-2)+u(t-3)$$
$$=5-6u(t-2)+u(t-3)$$

### Ejemplo

Escribir la función siguiente como una función a trozos y como combinación de funciones escalón unitaria




Solución

$$f(t)=\begin{cases}0, & 0\le t<4\\ 5, & t\ge 4\end{cases}$$

$$f(t)=0\left[u(t)-u(t-4)\right]+5u(t-4)$$

$$\boxed{f(t)=5u(t-4)}$$

**Ejemplo**

Escribir la función siguiente como una función a trozos y como combinación de funciones escalón unitaria




Solución

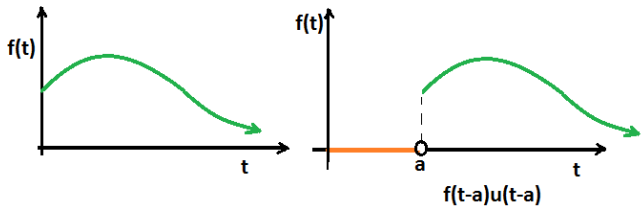
$$f(t)=\begin{cases}2t, & 0\le t<1\\ 0, & t\ge 1\end{cases}$$

$$f(t)=2t\left[u(t)-u(t-1)\right]+0u(t-1)$$

$$\boxed{f(t)=2t-2tu(t-1)}$$

**Traslación en el eje t**

$$f(t-a)u(t-a)=\begin{cases}0 & ,\ 0\le t<\ a\\ f(t-a) & ,\ t\ge a\end{cases}$$




**Teorema: Segundo teorema de traslación**

Si $F(s)=\mathscr{L}\{f(t)\}$ y $a>0$ entonces $\mathscr{L}\{f(t-a)\mathrm{u}(\mathrm{t}-\mathrm{a})\}=e^{-as}F(s)$

Un procedimiento para resolver estas transformadas de Laplace es el siguiente:

$$\mathscr{L}\{f(t-a)\mathrm{u}(\mathrm{t}-\mathrm{a})\} = e^{-as}\mathscr{L}\{f(t)\} = e^{-as}F(s)$$

En particular si $f(t)=1$, entonces

$$\mathscr{L}\{f(t-a)u(t-a)\} = \mathscr{L}\{u(t-a)\} = e^{-as}\mathscr{L}\{1\} = \frac{e^{-as}}{s}$$

**Ejemplo**

Evaluar $\mathscr{L}\left\{\cos\left[3(t-\frac{\pi}{2})\right]\mathrm{u}(\mathrm{t}-\frac{\pi}{2})\right\}$

Solución

Hacemos $f(t)=\cos(3t)$

$$\mathscr{L}\left\{\cos\left[3(t-\frac{\pi}{2})\right]\mathrm{u}(\mathrm{t}-\frac{\pi}{2})\right\} = e^{-\frac{\pi}{2}s}\mathscr{L}\{\cos(3t)\}$$

$$= e^{-\frac{\pi}{2}s}\frac{s}{s^2+9}$$

**Ejemplo:** Hallar $\mathscr{L}\{f(t)\}$ , si $f(t)=\begin{cases}5 & , \ 0\le t<2\\ -1 & , \ 2\le t<3\\ 0 & , \ t\ge 3\end{cases}$

Solución

Para no utilizar la definición de transformada de Laplace, primero expresamos la función seccionada en términos de las funciones escalón unitaria

$$f(t) = 5\left[u(t-0)-u(t-2)\right]-1\left[u(t-2)-u(t-3)\right]+0\left[u(t-3)\right]$$
$$= 5-5u(t-2)-u(t-2)+u(t-3)$$
$$= 5-6u(t-2)+u(t-3)$$
$$f(t) = 5-6u(t-2)+u(t-3)$$

Luego aplicamos la transformada de Laplace a esta función

$$\mathscr{L}\{f(t)\} = \mathscr{L}\{5-6u(t-2)+u(t-3)\}$$
$$= \mathscr{L}\{5\} - 6\mathscr{L}\{u(t-2)\} + \mathscr{L}\{u(t-3)\}$$
$$= 5\mathscr{L}\{1\} - 6e^{-2s}\mathscr{L}\{1\} + e^{-3s}\mathscr{L}\{1\}$$
$$F(s) = \frac{5}{s} - \frac{6e^{-2s}}{s} + \frac{e^{-3s}}{s}$$

**Ejemplo:** Hallar $\mathscr{L}\{f(t)\}$ si

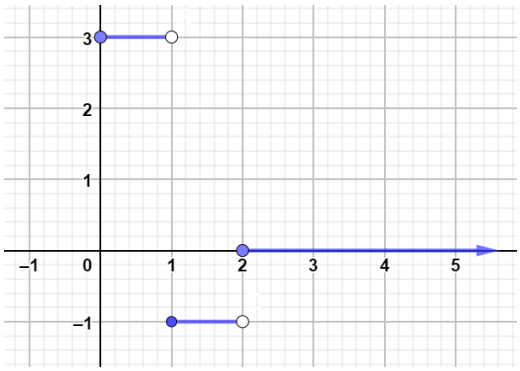
$$f(t) = \begin{cases} 1 & , \ 0 \le t < 1 \\ -t+2 & , \ 1 \le t < 2 \\ 0 & , \ 2 \le t < 3 \\ t-3 & , \ t \ge 3 \end{cases}$$

$$f(t) = 1\left[u(t) - u(t-1)\right] + (-t+2)\left[u(t-1) - u(t-2)\right] + 0\left[u(t-2) - u(t-3)\right] + (t-3)\left[u(t-3)\right]$$
$$f(t) = 1 - u(t-1) + (-t+2)u(t-1) - (-t+2)u(t-2) + (t-3)\left[u(t-3)\right]$$
$$f(t) = 1 + \left[-1 + (-t+2)\right]u(t-1) - (-t+2)u(t-2) + (t-3)\left[u(t-3)\right]$$
$$f(t) = 1 + \left[1-t\right]u(t-1) + (t-2)u(t-2) + (t-3)\left[u(t-3)\right]$$
$$f(t) = 1 - (t-1)u(t-1) + (t-2)u(t-2) + (t-3)u(t-3)$$

Luego la trasformada de Laplace es

$$\mathscr{L}\{f(t)\} = \mathscr{L}\{1 - (t-1)u(t-1) + (t-2)u(t-2) + (t-3)u(t-3)\}$$
$$\mathscr{L}\{f(t)\} = \mathscr{L}\{1\} - \mathscr{L}\{(t-1)u(t-1)\} + \mathscr{L}\{(t-2)u(t-2)\} + \mathscr{L}\{(t-3)u(t-3)\}$$
$$F(s) = \frac{1}{s} - e^{-s}\mathscr{L}\{t\} + e^{-2s}\mathscr{L}\{t\} + e^{-3s}\mathscr{L}\{t\}$$
$$F(s) = \frac{1}{s} - e^{-s}\left(\frac{1}{s^2}\right) + e^{-2s}\left(\frac{1}{s^2}\right) + e^{-3s}\left(\frac{1}{s^2}\right)$$
$$\boxed{F(s) = \frac{s - e^{-s} + e^{-2s} + e^{-3s}}{s^2}}$$

**Ejemplo:** Hallar la transformada de Laplace de la función que se muestra en la siguiente figura




Solución

$$f(t)=3\left[u(t)-u(t-1)\right]-1\left[u(t-1)-u(t-2)\right]+0u(t-2)$$

$$f(t)=3-4u(t-1)+u(t-2)$$

$$\mathscr{L}\left\{f(t)\right\}=\mathscr{L}\left\{3-4u(t-1)+u(t-2)\right\}$$

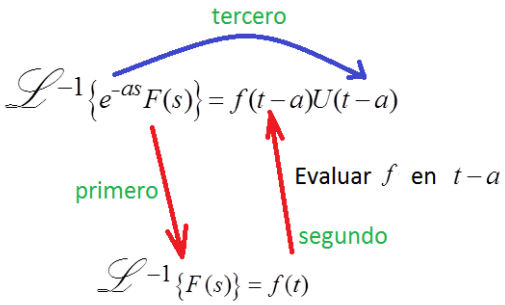
$$\mathscr{L}\left\{f(t)\right\}=3\mathscr{L}\left\{1\right\}-4\mathscr{L}\left\{u(t-1)\right\}+\mathscr{L}\left\{u(t-2)\right\}$$

$$\mathscr{L}\left\{f(t)\right\}=3\mathscr{L}\left\{1\right\}-4e^{-s}\mathscr{L}\left\{1\right\}+e^{-2s}\mathscr{L}\left\{1\right\}$$

$$\boxed{F(s)=\frac{3}{s}-\frac{4e^{-s}}{s}+\frac{e^{-2s}}{s}}$$

## Forma inversa del segundo teorema de traslación

Si $f(t)=\mathscr{L}^{-1}\{F(s)\}$ y $a>0$ entonces: $\mathscr{L}^{-1}\left\{e^{-as}F(s)\right\}=f(t-a)u(t-a)$



**Ejemplo:** Hallar $\mathscr{L}^{-1}\left\{\dfrac{se^{-\frac{\pi}{2}s}}{s^2+16}\right\}$

**Solución**

$$\mathscr{L}^{-1}\left\{\frac{se^{-\frac{\pi}{2}s}}{s^2+16}\right\}=\cos\left[4(t-\frac{\pi}{2})\right]u(t-\frac{\pi}{2})=\cos(4t-2\pi)u\left(t-\frac{\pi}{2}\right)=\cos(4t)u\left(t-\frac{\pi}{2}\right)$$

$\downarrow$ $\uparrow$ sustituimos $t$ por $t-\dfrac{\pi}{2}$

$$\mathscr{L}^{-1}\left\{\frac{s}{s^2+16}\right\}=\cos(4t)$$

$$\boxed{\mathscr{L}^{-1}\left\{\frac{se^{-\frac{\pi}{2}s}}{s^2+16}\right\}=\cos(4t)u\left(t-\frac{\pi}{2}\right)}$$

**Ejemplo:** Determine $\mathscr{L}^{-1}\left\{\dfrac{e^{-s}}{s^2+s}\right\}$

$$\mathscr{L}^{-1}\left\{\frac{e^{-s}}{s^2+s}\right\}$$

$$\mathscr{L}^{-1}\left\{\frac{1}{s^2+s}\right\}$$

$$\frac{1}{s^2+s}=\frac{1}{s(s+1)}$$

$$\frac{1}{s(s+1)}=\frac{A}{s}+\frac{B}{s+1}$$

$$1=A(s+1)+Bs$$

Si $s=0\rightarrow A=1$

Si $s=-1\rightarrow B=-1$

$$\mathscr{L}^{-1}\left\{\frac{1}{s^2+s}\right\}=\mathscr{L}^{-1}\left\{\frac{1}{s(s+1)}\right\}$$

$$=\mathscr{L}^{-1}\left\{\frac{1}{s}-\frac{1}{s+1}\right\}$$

$$=\mathscr{L}^{-1}\left\{\frac{1}{s}\right\}-\mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\}$$

$$\mathscr{L}^{-1}\left\{\frac{1}{s^2+s}\right\}=1-e^{-t}$$

$$\mathscr{L}^{-1}\left\{\frac{e^{-s}}{s^2+s}\right\}=\left(1-e^{-(t-1)}\right)u(t-1)$$

## Forma Alternativa del segundo teorema de traslación

$$\mathscr{L}\{g(t)u(t-a)\} = e^{-as}\mathscr{L}\{g(t+a)\}$$

**Ejemplo:** Encontrar $\mathscr{L}\{t^2u(t-2)\}$

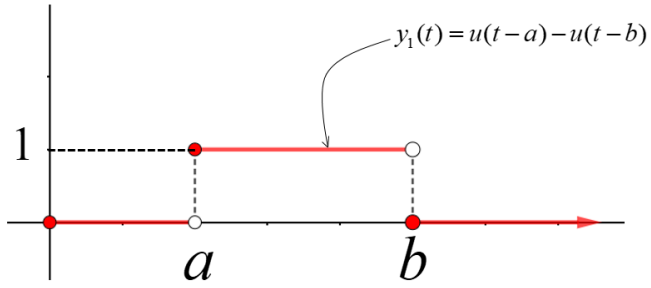
Solución

Hacemos $g(t) = t^2$ , $a = 2$

$$\begin{aligned}\mathscr{L}\{t^2u(t-2)\} &= e^{-2s}\mathscr{L}\{(t+2)^2\}\\ &= e^{-2s}\mathscr{L}\{t^2+4t+4\}\\ &= e^{-2s}\left[\mathscr{L}\{t^2\}+4\mathscr{L}\{t\}+4\mathscr{L}\{1\}\right]\\ &= e^{-2s}\left[\frac{2}{s^3}+\frac{4}{s^2}+\frac{4}{s}\right]\end{aligned}$$

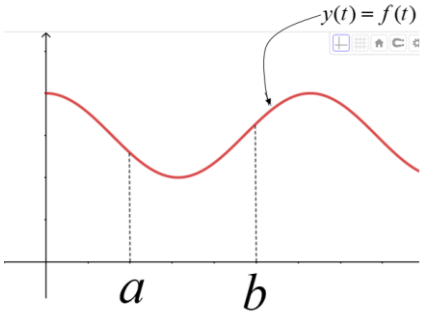
**Ejemplo**

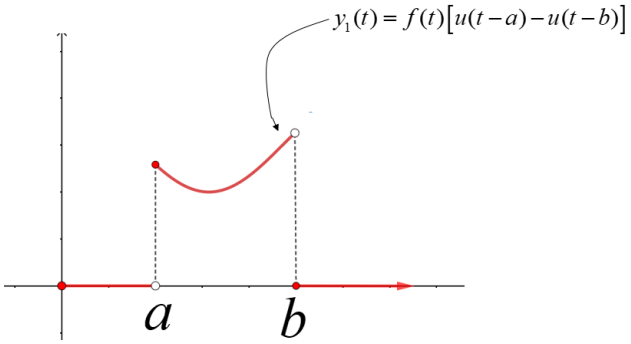
Hallar $\mathscr{L}\{[f(t)][y_1(t)]\}$, donde $y_1(t) = u(t-a)-u(t-b), \quad a<b$ (conocida como función <<filtro>>)




$$y_1(t) = u(t-a)-u(t-b) = \begin{cases}0, & 0 \le t < a\\ 1, & a \le t < b\\ 0, & t \ge b\end{cases}$$

Efecto de corte




$$y = f(t)\left[u(t-a)-u(t-b)\right] = \begin{cases} 0, & 0 \le t < a \\ f(t), & a \le t < b \\ 0, & t \ge b \end{cases}$$

$$\mathscr{L}\{f(t)\} = \mathscr{L}\{f(t)\left[u(t-a)-u(t-b)\right]\}$$
$$\mathscr{L}\{f(t)\} = \mathscr{L}\{f(t)u(t-a)\} - \mathscr{L}\{f(t)u(t-b)\}$$
$$\mathscr{L}\{f(t)\} = e^{-as}\mathscr{L}\{f(t+a)\} - e^{-bs}\mathscr{L}\{f(t+b)\}$$

**Teorema: (Derivadas de transformadas)**

Si $F(s) = \mathscr{L}\{f(t)\}$ y si n=1,2,3,..., entonces:

$$\mathscr{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}\mathscr{L}\{f(t)\}$$
$$\mathscr{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}F(s)$$

**Ejemplo:** hallar $\mathscr{L}\{t^2 senh(t)\}$

$$\mathscr{L}\{t^2 senh(t)\} = (-1)^2 \frac{d^2}{ds^2}\mathscr{L}\{senh(t)\}$$

$$= \frac{d}{ds}\left[\frac{d}{ds}\left(\frac{1}{s^2-1}\right)\right]$$

$$= \frac{d}{ds}\left(\frac{-2s}{(s^2-1)^2}\right)$$

$$= \frac{-2(s^2-1)^2 + 2s(2)(s^2-1)2s}{(s^2-1)^4}$$

$$= \frac{-2(s^2-1)\left[(s^2-1)-2s2s\right]}{(s^2-1)^4}$$

$$= \frac{-2\left[s^2-1-4s^2\right]}{(s^2-1)^3}$$

$$= \frac{-2(-3s^2-1)}{(s^2-1)^3}$$

$$\mathscr{L}\{t^2 senh(t)\} = \frac{2(3s^2+1)}{(s^2-1)^3}$$

**Ejemplo:** hallar $\mathscr{L}^{-1}\left\{\ln\left(\frac{s-3}{s+1}\right)\right\}$

Solución

$$F(s) = \ln\left(\frac{s-3}{s+1}\right) = \ln(s-3) - \ln(s+1)$$

$$-\frac{d}{ds}[F(s)] = -\frac{d}{ds}[\ln(s-3) - \ln(s+1)] = \frac{1}{s+1} - \frac{1}{s-3}$$

$$\mathscr{L}\{t\ f(t)\} = -\frac{d}{ds}[F(s)] = \frac{1}{s+1} - \frac{1}{s-3}$$

$$t\ f(t) = \mathscr{L}^{-1}\left\{\frac{1}{s+1} - \frac{1}{s-3}\right\}$$

$$t\ f(t) = e^{-t} - e^{3t}$$

$$\boxed{f(t) = \frac{e^{-t} - e^{3t}}{t}}$$